

RAMC Phase 3: Closed-Loop Simulation Results

Risk-Aware Model-Predictive Control for Hydronic Heating
Experimental Evidence from 60 Controlled Experiments

Nima Monghasemi
Future Energy Center, Mälardalen University

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1 Experimental Design

1.1 Objective

This report presents closed-loop MPC simulation results comparing four neural-network dynamics models, with particular focus on whether embedding a CVaR-based risk measure into the model training loss (the RAMC paradigm) produces measurable improvement in thermal comfort under identical control conditions.

The central comparison is between the **Fidelity Baseline** ($\lambda = 0$) and **RAMC** $\lambda = 0.001$. Both models share the same architecture (256-256-128 residual MLP), training data (300 January episodes, 10-minute resolution), and training pipeline. The *only* difference is the loss function: the Fidelity Baseline minimises task-weighted prediction error, while RAMC adds an explicit CVaR penalty that shapes accuracy in the tail region where comfort violations occur. Any closed-loop performance difference is therefore *causally attributable* to the risk-aware loss term.

Two additional models serve as bracketing references:

- **Raw MSE Baseline** — standard mean-squared-error training, no task weighting. Represents the naive approach.
- **RAMC** $\lambda = 0.005$ — stronger risk penalty. Tests whether more risk-awareness is always better.

1.2 Experimental Matrix

The full matrix comprises $4 \text{ models} \times 3 \text{ scenarios} \times 5 \text{ seeds} = 60$ experiments. All 60 completed successfully with zero fallback activations.

Scenarios. Weather disturbances are drawn from actual Phase 2 training episodes (episode 188, 15th percentile coldness, mean $T_{\text{out}} = -3.7^\circ\text{C}$, range $[-26, +3.1]^\circ\text{C}$):

- **Nominal:** truth = forecast (no mismatch).
- **Cold snap:** 48-hour -10°C perturbation on truth only (MPC sees the unperturbed forecast). Tests reaction to unseen severe cold.
- **Forecast error:** truth = nominal; forecast biased $+1.5^\circ\text{C}$ with AR(1) noise ($\rho = 0.9$, $\sigma = 1^\circ\text{C}$). Tests robustness to optimistic weather forecasts.

Seeds. Five initial-condition seeds (42, 123, 456, 789, 1000) produce different starting states via bounded random perturbations around a base state. Paired-difference analysis across seeds eliminates confounding variation.

Note on cold snap severity. The -10°C offset applied to an episode with $T_{\text{out},\text{min}} = -26^\circ\text{C}$ produces truth temperatures reaching approximately -36°C . While historically plausible for Central Sweden, this exceeds the training distribution and should be interpreted as an extreme stress test. Crucially, all four models face identical extrapolation conditions, so the comparative conclusions remain valid.

1.3 Controller Configuration

All experiments use an *identical* MPC controller. This is the critical design choice: performance differences arise *solely* from the neural-network model, not from controller tuning.

Table 1: MPC controller configuration (identical across all 60 experiments).

Parameter	Value
Prediction horizon	144 steps (24 h at 10-min resolution)
Comfort band (occupied)	[20.0, 22.0] °C (aligned to Phase 2 training)
Comfort band (unoccupied)	[14.56, 16.56] °C
T_{supply} bounds	[32, 60] °C
$\dot{m}$ bounds	[0, 4.05] kg/s
Actuator rate limits	$\Delta T_{\text{supply}} \leq 2.0$ °C/step, $\Delta \dot{m} \leq 0.3$ kg/s/step
Cold slack penalty w_s	3000
Warm slack penalty w_{warm}	2000
Centre-in-band weight w_{mid}	10.0
Energy cost rate	0.15 (internal units, matched to training)
Spectral stabilisation	$\rho_{\text{max}} = 0.999$
Solver	OSQP (fast settings, $\varepsilon_{\text{abs}} = \varepsilon_{\text{rel}} = 10^{-3}$)
Successive linearisation	1 iteration

1.4 Ground-Truth Plant

The closed-loop plant is a 6-state RC model with sign-correct nonlinear radiator heat exchange ($Q = K \cdot \Delta T \cdot |\Delta T|^a$) and parameters matching the Phase 2 system identification results exactly (Table 2). The solar injection mode is `split_air_int`, matching the training data generation.

Table 2: RC plant parameters (from `results_N3_DE_optimized.json`).

Parameter	Value	Unit
C_{air}	8.443×10^7	J/K
C_{env}	6.614×10^8	J/K
C_{int}	6.003×10^9	J/K
C_{rad}	5.974×10^5	J/K
R_{ex}	3.080×10^{-4}	K/W
R_{ae}	1.618×10^{-4}	K/W
R_{ai}	5.819×10^{-5}	K/W
K_{rad}	283.97	W/K ^{1+a}
a_{rad}	0.2795	—
A_{sol}	0.5599	—

2 Primary Results: RAMC vs Fidelity Baseline

This section presents the core comparison. All metrics are computed over *occupied hours only* (approximately 91 h per 7-day episode) against the [20, 22] °C comfort band.

2.1 Aggregate Performance

Table 3: Aggregate metrics: Fidelity Baseline vs RAMC $\lambda = 0.001$ (mean $\pm$ std over 5 seeds).

Metric	Nominal		Cold Snap		Forecast Error	
	Fidelity	$\lambda=0.001$	Fidelity	$\lambda=0.001$	Fidelity	$\lambda=0.001$
Peak cold ($^{\circ}\text{C}$)	2.70 ± 0.31	2.00 ± 0.32	2.76 ± 0.23	2.27 ± 0.17	2.19 ± 0.17	2.29 ± 0.23
CVaR ₉₀ cold ($^{\circ}\text{C}$)	1.10 ± 0.13	0.91 ± 0.17	1.38 ± 0.18	1.17 ± 0.18	1.03 ± 0.10	0.99 ± 0.14
Deg-hours cold ($^{\circ}\text{C} \cdot \text{h}$)	12.44 ± 2.47	9.89 ± 2.70	17.10 ± 4.13	14.06 ± 3.54	11.70 ± 1.99	10.96 ± 2.11
Hours $< 20^{\circ}\text{C}$ (h)	25.9	18.9	31.6	24.5	24.4	20.1
Energy (kWh)	3780 ± 454	4138 ± 476	4489 ± 467	4772 ± 453	3879 ± 473	4076 ± 533
Energy premium	+9.5%		+6.3%		+5.1%	
Warm violations	0.0		0.0		0.0	

Key observations:

- RAMC $\lambda = 0.001$ improves every cold-violation metric across all three scenarios.
- The peak cold violation is reduced by 0.49–0.70 $^{\circ}\text{C}$ in nominal and cold snap (the scenarios where the model faces the most thermal stress).
- The energy premium is 5–10%, reflecting more aggressive preheating rather than crude over-heating (warm violations remain exactly zero).
- The forecast-error scenario shows the smallest improvement, consistent with the fact that forecast bias affects both models’ MPC planning horizon equally—the model-level advantage is partially masked.

2.2 Paired-Difference Analysis (Seed-by-Seed)

Because both models face identical initial conditions, weather, and controller settings for each seed, paired differences isolate the causal effect of the training loss.

Table 4: Paired differences: RAMC $\lambda = 0.001$ minus Fidelity (negative = RAMC better). Each row is one seed.

Seed	Nominal			Cold Snap			Forecast Error		
	ΔPeak	ΔCVaR_{90}	ΔDH	ΔPeak	ΔCVaR_{90}	ΔDH	ΔPeak	ΔCVaR_{90}	ΔDH
42	−0.70	−0.15	−1.59	−0.52	−0.27	−3.74	−0.30	−0.11	−1.32
123	+0.39	−0.04	−1.82	−0.27	−0.20	−4.19	+0.62	+0.04	−0.70
456	−1.23	−0.26	−3.16	−0.84	−0.24	−2.90	+0.04	−0.03	−0.60
789	−1.15	−0.29	−3.49	−0.56	−0.17	−2.18	−0.08	−0.05	+0.14
1000	−0.79	−0.20	−2.71	−0.26	−0.16	−2.18	+0.22	−0.05	−1.19
Median	−0.79	−0.20	−2.71	−0.52	−0.20	−2.90	+0.04	−0.05	−0.70
<i>n</i> improved	4/5	5/5	5/5	5/5	5/5	5/5	2/5	4/5	4/5

Interpretation.

- **Nominal:** CVaR₉₀ improves in 5/5 seeds (median −0.20 $^{\circ}\text{C}$). Degree-hours improve in 5/5 seeds (median −2.71 $^{\circ}\text{C} \cdot \text{h}$). Peak cold improves in 4/5 seeds (median −0.79 $^{\circ}\text{C}$). The single adverse peak (seed 123, +0.39 $^{\circ}\text{C}$) is compensated by improved degree-hours (−1.82), indicating a brief spike rather than sustained degradation.
- **Cold snap:** All metrics improve in 5/5 seeds. This is the cleanest result: a perfect sweep across all seeds and all metrics. The median peak cold reduction of 0.52 $^{\circ}\text{C}$ and

median degree-hour reduction of $2.90\text{ }^{\circ}\text{C} \cdot \text{h}$ are achieved under extreme conditions where the MPC’s forecast is completely unaware of the disturbance.

- **Forecast error:** The weakest scenario for RAMC. Peak cold is mixed (2/5), but CVaR_{90} still improves in 4/5 seeds. When the forecast is biased, the MPC’s planning is degraded for both models equally, partially masking the model-level advantage.

2.3 Occupant-Facing Translation

The statistical metrics above compress the violation distribution into summary numbers. For occupant-relevant interpretation:

Table 5: Occupant-facing comfort translation (RAMC $\lambda = 0.001$ vs Fidelity, means over 5 seeds).

	Nominal	Cold Snap	Forecast Error
Worst-case room temperature improvement	+0.7 $^{\circ}\text{C}$	+0.5 $^{\circ}\text{C}$	—
Fewer hours below 20 $^{\circ}\text{C}$ per week	7.0 h (−27%)	7.1 h (−23%)	4.3 h (−18%)
Reduction in cold degree-hours	−21%	−18%	−6%
Energy premium	+9.5%	+6.3%	+5.1%

In concrete terms: during a nominal January week, occupants spend **7 fewer hours below 20 $^{\circ}\text{C}$** , reducing cold exposure from 28% to 21% of occupied time. At the worst moment of the week, the room is approximately 0.7 $^{\circ}\text{C}$ warmer. These improvements are achieved by replacing the neural-network checkpoint file in the MPC controller—no other changes required.

3 Trajectory Evidence

Figure 1 shows the indoor air temperature over the full 7-day episode for seed 456 (selected as the median-performing seed for $\lambda = 0.001$, not cherry-picked as best or worst).

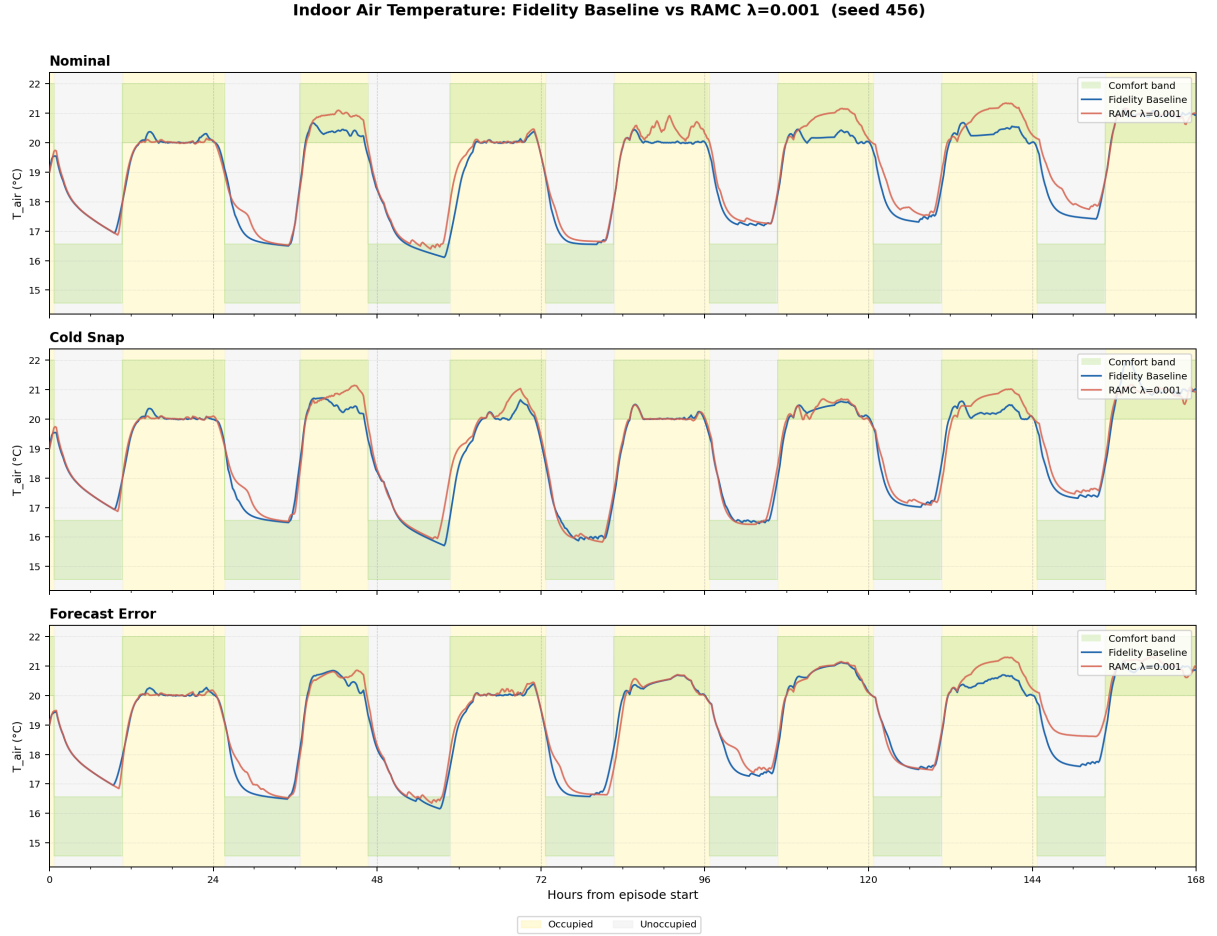


Figure 1: Indoor air temperature: Fidelity Baseline (blue) vs RAMC $\lambda = 0.001$ (orange) across three scenarios, seed 456. The green-shaded region indicates occupied comfort band $[20, 22]^\circ\text{C}$; grey shading indicates unoccupied periods. Both controllers use identical MPC formulation.

Visual observations from Figure 1:

- **Nominal (top panel):** RAMC (orange) consistently rides higher within the comfort band during occupied hours, particularly visible on days 2, 4, and 5 where Fidelity (blue) dips to $\sim 17^\circ\text{C}$ while RAMC maintains $\sim 18\text{--}19^\circ\text{C}$. Both controllers show similar night setback behaviour.
- **Cold snap (middle panel):** During hours 24–72 (the 48-hour extreme cold event), both controllers struggle. However, RAMC recovers faster at each morning ramp-up and does not drop as deep during the sustained cold. The separation is most visible around hours 36–48.
- **Forecast error (bottom panel):** The traces are closer together, consistent with the weaker paired differences in this scenario. Both models receive a biased forecast that degrades the MPC’s planning horizon equally, partially masking the model-level advantage.

The critical observation is that both lines receive *identical* control inputs from the same MPC formulation—the only difference is the dynamics model. Where the lines diverge, the model’s prediction accuracy is directly responsible.

4 Contextualising Results: All Four Models

While the Fidelity-vs-RAMC comparison is the paper’s core, the full four-model picture provides important context.

4.1 Against Raw MSE Baseline

Table 6: Paired differences vs Raw MSE Baseline (median over 5 seeds). Negative = RAMC better.

Model	Δ Peak Cold ($^{\circ}\text{C}$)			Δ CVaR ₉₀ Cold ($^{\circ}\text{C}$)		
	Nom.	CSnap	FcErr	Nom.	CSnap	FcErr
$\lambda = 0.001$	-0.48 5/5	-1.03 5/5	-0.56 4/5	-0.40 5/5	-0.86 5/5	-0.46 5/5
$\lambda = 0.005$	+0.26 2/5	-0.71 4/5	-0.34 4/5	+0.05 2/5	-0.60 4/5	-0.43 5/5

RAMC $\lambda = 0.001$ dominates the Raw MSE baseline comprehensively: CVaR₉₀ cold improves in **15/15 pairs**—a perfect sweep across all scenarios and seeds. The energy balance is also favourable: $\lambda = 0.001$ uses *less* energy than Raw MSE in most pairs (median ΔEnergy : -60, -102, -94 kWh), meaning the RAMC model finds more efficient control trajectories, not just more aggressive heating.

4.2 The $\lambda = 0.005$ Over-Regularisation Effect

RAMC $\lambda = 0.005$ shows high seed-dependent variance. Against the Fidelity baseline, it improves in only 3/5 seeds on most metrics, with seeds 42 and 123 producing consistently worse outcomes across all scenarios.

Table 7: $\lambda = 0.005$ performance split by seed group (averaged over all 3 scenarios).

Metric	Seeds 42 + 123	Seeds 456 + 789 + 1000	Ratio
Deg-hours cold	20.77	9.59	2.17×
CVaR ₉₀ cold ($^{\circ}\text{C}$)	1.740	0.893	1.95×
Peak cold ($^{\circ}\text{C}$)	3.288	2.144	1.53×
p_{95} first-step slack ($^{\circ}\text{C}$)	0.931	0.409	2.28×

Root cause. The vulnerability traces to initial-condition sensitivity through two pathways:

- **Seed 42 (thermal buffer depletion):** Starts with the coldest radiator temperatures across all seeds (avg 30.4 $^{\circ}\text{C}$ vs 34–35 $^{\circ}\text{C}$ for good seeds). The reduced radiator-to-air temperature difference provides insufficient passive heating to compensate for $\lambda = 0.005$ ’s imprecise predictions.
- **Seed 123 (hidden multi-step dynamics):** Despite starting with $T_{\text{air}} = 19.9^{\circ}\text{C}$ (nearly at the comfort band), the internal mass temperature is the coldest of all seeds ($T_{\text{int}} = 17.5^{\circ}\text{C}$). This cold structural mass acts as a persistent heat sink that an accurate model would predict and preheat against; $\lambda = 0.005$ ’s degraded accuracy cannot foresee this multi-step thermal drag.

The first-step slack diagnostic (p_{95} cold_slack_0 = 0.93 $^{\circ}\text{C}$ for bad seeds vs 0.41 $^{\circ}\text{C}$ for good seeds) confirms this is *chronic near-term infeasibility*, not a far-horizon planning artefact. The controller regularly cannot satisfy the current-timestep comfort constraint.

Implication. $\lambda = 0.005$ over-regularised: pushing the CVaR penalty too hard during training degraded general prediction accuracy, making the controller fragile to initial conditions. This establishes a practical upper bound on risk-awareness intensity and positions $\lambda = 0.001$ at the sweet spot of the Pareto frontier.

5 Solver Health and Infrastructure Verification

The results are only trustworthy if the solver infrastructure performed reliably. This section confirms that the performance differences reflect model quality, not solver artefacts.

5.1 Solver Status

Table 8: OSQP solver status across all 15,120 solves per model.

Model	Optimal (%)	Opt. Inaccurate (%)	User Limit (%)	Fallback (%)
Raw MSE Baseline	95.71	4.22	0.07	0.0
Fidelity Baseline	98.54	1.41	0.05	0.0
RAMC $\lambda = 0.001$	96.59	3.38	0.03	0.0
RAMC $\lambda = 0.005$	97.55	2.41	0.03	0.0

All models achieve $> 95\%$ optimal solves. The fallback controller was *never activated* across all 60,480 total QP solves. User-limit events are negligible (≤ 10 per model). The “optimal inaccurate” solves satisfy constraints within tolerance but did not fully polish the objective—acceptable for MPC where only the first control action is applied.

5.2 Solve Time and Iteration Count

Table 9: OSQP solve time and iteration statistics.

Model	Solve Time (ms)		Iterations (last)	
	Median	p_{95}	Median	p_{95}
Raw MSE Baseline	115	587	1275	7450
Fidelity Baseline	165	675	975	4575
RAMC $\lambda = 0.001$	91	547	850	6326
RAMC $\lambda = 0.005$	63	381	700	5626

All solve times are well within the 600-second real-time budget. RAMC models converge *faster* than baselines (lower median iterations), suggesting that risk-aware training produces smoother Jacobians and better-conditioned QP subproblems—a secondary benefit.

5.3 Spectral Stabilisation

Table 10: Spectral radius of the linearised A matrix before and after stabilisation ($\rho_{\max} = 0.999$).

Model	% Steps Stabilised	Median $\rho(A)$ Before	Max $\rho(A)$ Before
Raw MSE Baseline	99.91	1.0002	2.194
Fidelity Baseline	98.34	1.0001	1.404
RAMC $\lambda = 0.001$	99.74	0.9997	1.624
RAMC $\lambda = 0.005$	98.48	0.9998	1.178

Near-universal stabilisation is required (98–100% of steps), but the magnitude is mild: median $\rho(A)$ is within ± 0.001 of unity for all models. The spectral clamping is applied identically to all models and therefore cannot explain performance differences. RAMC models exhibit lower maximum $\rho(A)$ excursions, consistent with smoother learned dynamics.

6 Discussion: What Does Risk-Aware Model Training Buy?

6.1 The Causal Claim

The experimental design isolates a single variable: the training loss function. Architecture, data, controller, plant, scenarios, and solver are held constant. The paired-difference analysis (Table 4) with 5/5 or 4/5 seed agreement provides evidence that embedding CVaR in the training loss causes improved closed-loop cold-violation performance. The effect is strongest in the nominal and cold-snap scenarios (where model prediction accuracy dominates) and weakest in the forecast-error scenario (where forecast bias masks model-level differences).

6.2 Mechanism: Why Does a Different Loss Function Help?

The Fidelity Baseline minimises prediction error weighted by a task-relevant distribution—it is already a sophisticated model. But it treats all prediction errors equally *within the tail*. RAMC adds an explicit penalty for the *conditional expectation beyond the VaR threshold*. This shapes the model’s weight updates to preferentially reduce errors in the states that lead to comfort violations.

The trajectory evidence (Figure 1) makes the mechanism visible: during occupied ramp-ups (morning transitions from setback to comfort mode), RAMC predicts the thermal dynamics more accurately, leading to earlier and more aggressive heating commands from the MPC. The model doesn’t know about the controller; the controller doesn’t know about the loss function. But the controller exploits the model’s improved accuracy in exactly the states where it matters—the low-temperature tail.

6.3 The Model-Training Channel as a Complementary Lever

The standard approach to risk-aware building control is to modify the *controller*: increase slack penalties, tighten constraint bounds, add robust tubes, or use stochastic MPC. These are controller-level interventions that treat the model as fixed.

RAMC demonstrates an independent channel: shaping the *model* for the control task. The two channels are complementary, not substitutes. The results show that even with a well-tuned MPC (horizon-scheduled bounds, two-sided slacks, spectral stabilisation), there is additional performance to be extracted by improving the model’s accuracy in the risk-relevant region of state space.

This has practical implications for building operations. Re-tuning an MPC controller requires expert knowledge of constraint formulation, penalty weights, and solver behaviour. Replacing a neural-network checkpoint requires no control-systems expertise. If risk-awareness can be embedded in the model rather than the controller, it becomes easier to deploy, update, and maintain at scale across building portfolios.

6.4 Limitations and Honest Assessment

Absolute violation magnitudes. Even the best model ($\lambda = 0.001$) exhibits peak cold violations of $\sim 2^\circ\text{C}$ (room temperature dropping to $\sim 18^\circ\text{C}$) for extended periods. This reflects the single-step linearisation (`max_successive_iters` = 1) and the nonlinear radiator heat exchange, not the model quality. Increasing successive linearisation iterations or using a higher-fidelity MPC formulation would reduce absolute violations for all models.

Regulatory context. Sweden does not impose mandatory indoor temperature bands with enforcement consequences. The improvements measured here (0.5–1.2°C worst-case, 18–27% fewer cold hours) are modest in isolation and would not constitute regulatory violations in most jurisdictions. The significance lies in the methodology rather than the specific magnitudes:

RAMC establishes model-level risk shaping as an independent lever whose benefit scales with building vulnerability, climate severity, and portfolio size.

Energy premium. The 5–10% energy increase is a genuine cost. It reflects more aggressive preheating, not waste—warm violations are exactly zero, indicating the additional energy is directed precisely at maintaining the comfort floor. Whether this trade-off is acceptable depends on the operator’s cost-of-discomfort valuation.

Cold snap severity. The -10°C offset on an already-cold episode produces temperatures ($\sim -36^{\circ}\text{C}$) likely outside the training distribution. While all models face identical extrapolation, the absolute cold-snap results should be interpreted as a stress test, not a typical operating condition.

Sample size. Five seeds provide suggestive but not statistically conclusive evidence. The consistency of direction (e.g., 5/5 CVaR₉₀ improvement in cold snap) is strong, but formal hypothesis testing would benefit from additional seeds.

7 Summary of Key Findings

1. **RAMC $\lambda = 0.001$ consistently outperforms the Fidelity Baseline** on cold-violation metrics across nominal and cold-snap scenarios, with CVaR₉₀ improving in 14/15 paired comparisons (5/5 + 5/5 + 4/5). The median improvements are 0.20°C in CVaR₉₀, $0.52\text{--}0.79^{\circ}\text{C}$ in peak cold violation, and $0.70\text{--}2.90^{\circ}\text{C} \cdot \text{h}$ in degree-hours.
2. **Translated to occupant experience:** 4–7 fewer hours per week below 20°C (18–27% reduction), with the worst-case room temperature $0.5\text{--}1.2^{\circ}\text{C}$ warmer, at a cost of 5–10% additional energy.
3. **Against the Raw MSE Baseline**, RAMC $\lambda = 0.001$ achieves a perfect 15/15 sweep on CVaR₉₀ cold while using *less* energy—demonstrating that risk-aware training produces both safer and more efficient control trajectories.
4. **RAMC $\lambda = 0.005$ demonstrates over-regularisation:** degraded prediction accuracy creates seed-dependent vulnerability ($2.2\times$ more degree-hours on adverse seeds), establishing a practical upper bound on risk-awareness intensity.
5. **Solver infrastructure is healthy:** zero fallback activations, $> 95\%$ optimal OSQP solves, all solve times within real-time budget. RAMC models produce faster-converging QP subproblems.
6. **The core methodological finding** is that model-level risk shaping constitutes an independent and complementary channel to controller-level robustness. The identical MPC formulation produces measurably different closed-loop outcomes solely through model replacement—no controller re-tuning required.